

Putting plural definites into context*

Jacopo Romoli

Heinrich Heine Universität Düsseldorf

Yasutada Sudo

University College London

Petra Augurzky

Goethe-Universität Frankfurt

Marion Bonnet

Georg-August-Universität Göttingen

Richard Breheny

University College London

Alexandre Cremers

Vilniaus Universitetas

Cornelia Ebert

Goethe-Universität Frankfurt

Kurt Erbach

Goethe-Universität Frankfurt

Markus Steinbach

Georg-August-Universität Göttingen

Clemens Mayr

Georg-August-Universität Göttingen

Abstract There is an ongoing empirical debate concerning the availability of *non-maximal readings* of plural definites in semantically negative grammatical environments. Some theories of definite plurals (Križ 2015, 2016, Križ & Spector 2021, among others) assume that, all else being equal, non-maximal readings are equally available in positive and negative environments, whereas other approaches maintain that such readings are unavailable or less readily available in negative environments (Magri 2014, Bar-Lev 2018, 2021, Guerrini & Wehbe 2025). What makes this debate difficult to resolve on the basis of introspective judgments is that non-maximal readings are inherently context-sensitive, and positive and negative sentences, even when they are truth-conditionally equivalent, are typically used in different types of conversational contexts. As a result, direct comparisons between sentence of different polarity are often inconclusive. This paper reports three experiments investigating how the availability of non-maximal readings is affected by two distinct background contexts in the scope of four quantificational expressions: *every*, *no*, *not every*, and *exactly two*. Our results show that contextual manipulation has a robust effect for definite plurals under *every* and *not every*, but a substantially weaker effect under *no* and *exactly two*. We argue that these findings pose a challenge for both of the aforementioned approaches to non-maximality.

Keywords: plural definites, homogeneity, non-maximality, context sensitivity, relevance

* We would like to thank TBA. The first two experiments presented below are also published in Augurzky et al. 2023. The current paper builds on that initial work by adding a third experiment and a detailed discussion of the overall results with respect to the two main approaches in the literature.

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1 Introduction

Sentences containing a plural definite expression such as *his presents* in (1a) exhibit two distinctive interpretive properties: *homogeneity* and *non-maximality* (see Križ 2019 for an overview). Homogeneity concerns the intuition that the individuals constituting the maximal plurality denoted by the definite are evaluated uniformly with respect to predication: they either all make it true or all make it false. In particular, while (1a) is typically interpreted similarly to its universal counterpart in (1b), its negation in (2a) is more naturally paraphrased by (2b) than by the negation of (1b), as one might otherwise expect.

- (1) a. Frank opened his presents.
b. Frank opened all of his presents.
- (2) a. Frank didn't open his presents.
b. Frank didn't open any of his presents.

For different approaches to homogeneity, see Schwarzschild 1994, Löbner 2000, Gajewski 2005, Križ 2016, Bar-Lev 2021, Magri 2014, Križ & Spector 2021, Sbardolini 2023, among others.

Non-maximality, by contrast, refers to a particular form of context sensitivity. Depending on the conversational context, the truth conditions of sentences containing plural definites are perceived to be less rigid than those of sentences containing quantifiers. For instance, (1a), unlike (1b), can be judged true in certain contexts even if Frank did not open all of his presents. A context that intuitively licenses such an interpretation is one in which it matters whether Frank opened even one of his presents. We return below to the kinds of contexts that give rise to non-maximality. For further discussion, see Krifka 1996, Lasersohn 1999, Malamud 2012, Križ 2016, Križ & Spector 2021, Bar-Lev 2018, 2021, Guerrini & Wehbe 2025, among others.

There is currently an empirical debate concerning the availability of non-maximal readings for definite plurals in negative environments such as (2a), and competing theories make divergent predictions.¹

Some theories (Križ 2015, 2016, Križ & Spector 2021, among others) assume that, all else being equal, non-maximal readings should be equally available in

¹ More precisely, the relevant distinction corresponds to upward and downward entailing environments. A function f of type $\langle \gamma, \tau \rangle$ is upward entailing (UE) iff for all x, y of type γ such that $x \Rightarrow y$, $f(x) \Rightarrow f(y)$:

(i) A function f is UE iff $\forall x, y (x \Rightarrow y \rightarrow f(x) \Rightarrow f(y))$.

A function is downward entailing (DE) iff it licenses entailments from supersets to subsets:

(ii) A function f is DE iff $\forall x, y (x \Rightarrow y \rightarrow f(y) \Rightarrow f(x))$.

positive and negative environments. We refer to these as *symmetric theories*. By contrast, others (Magri 2014, Bar-Lev 2018, 2021, Guerrini & Wehbe 2025, among others) maintain that non-maximal readings are unavailable, or at least substantially less available, in negative environments. We call these *asymmetric theories*.

We return to the details of these approaches below, but it is useful to briefly indicate why they diverge in their predictions. In symmetric theories, homogeneity involves truth-value gaps or underspecification, and non-maximality arises from the interaction between these gaps and contextual factors. Since neither ingredient is sensitive to polarity, non-maximal readings are predicted to be equally available in positive and negative environments. Asymmetric theories, by contrast, analyze homogeneity and non-maximality in terms of implicatures and their contextual modulation. In the version we focus on here, implicatures are presuppositional in nature and can project and be accommodated. As we will see, this yields asymmetric predictions.

A major obstacle to resolving this debate is the subtlety of context sensitivity and the complexity of the empirical landscape. Moreover, positive and negative sentences are typically used in different conversational contexts (see Horn 1989, Tian & Breheny 2019). As a result, differences in interpretation may reflect contextual preferences rather than genuine semantic asymmetries.

To illustrate the relevant contextual effects, consider again the positive and negative sentences with plural definites above, (1a) and (2a), in a what we can call a ‘non-maximal scenario’ in which Frank opens some but not all of his presents. And compare the following contexts.

- (3) **The Prohibited context:** Frank’s family has a rule that he must wait for his grandparents before opening presents. Opening presents beforehand is prohibited. (Implicit QUD: Did Frank open any of his presents?)
- (4) **The Required context:** Frank’s family requires him to open his presents before his aunt and uncle arrive. Opening presents beforehand is required. (Implicit QUD: Did Frank open all of his presents?)

As we will explain below, in both approaches, (1a) is more likely to be judged true in the Prohibited context than in the Required one. This contrast does not arise for the corresponding quantified sentence in (1b). Both theories therefore make the same prediction for positive sentences. They diverge, however, with respect to negative sentences. Symmetric theories predict a parallel contextual effect in the opposite direction: (2a) should be more acceptable in the Required context. Asymmetric

A function is non-monotonic if it is neither UE nor DE. For present purposes, *every* is upward entailing, *no* and *not every* are downward entailing, and *exactly two* is non-monotonic. We continue to refer to these environments as positive, negative, and non-monotonic, respectively.

theories, by contrast, predict no contextual effect: (2a) should be judged false in both contexts.

Previous studies have not systematically controlled for the potential confound introduced by differences in conversational context. This includes experimental work such as Križ & Chemla 2015 and Tieu, Križ & Chemla 2019. In particular, Križ & Chemla 2015 introduced two methodological innovations, which we have adopted in our studies. First, they employed intermediate judgment scales to probe non-categorical responses. Second, they investigated quantified sentences in order to control scope.

- (5) a. Every boy opened his presents.
- b. No boy opened his presents.

Their results show that all target sentences elicited more intermediate judgments than control sentences, indicating the availability of non-maximal readings. They also observed more intermediate judgments in positive than in negative sentences, a pattern that appears compatible with asymmetric theories. However, since their experiments did not specify conversational contexts beyond the visual scenes, participants may have imagined different contexts for different sentence types, potentially driving this asymmetry.

Similarly, Tieu, Križ & Chemla report asymmetric patterns in both children and adults. However, their experiments provided limited contextual information and did not control for scope, leaving open alternative explanations.

Taken together, previous findings do not provide decisive evidence for either approach. In this paper, we report three experiments that examine how truth-value judgments vary across systematically manipulated conversational contexts. We build on Križ & Chemla's aforementioned innovations by using intermediate response scales and quantified sentences with bound pronouns.

- (6) a. Every boy opened his presents.
- b. No boy opened his presents.
- c. Not every boy opened his presents.
- d. Exactly two boys opened their presents.

The cases involving *no*, *not every*, and *exactly two* are particularly informative, since they generate divergent predictions across theories and have not been systematically tested before.

To preview our main findings: definite plurals can receive non-maximal readings in all tested environments. However, sensitivity to contextual manipulation varies in ways that challenge both approaches. We observe an asymmetry between *every* and *no*, consistent with asymmetric theories, but also parallel effects for *every*, *not*

every, and both components of *exactly two*, consistent with symmetric theories. Our findings, therefore, are challenging for both approaches.

The remainder of the paper is organized as follows. Section 2 reviews the theoretical background. Sections 3–5 present the experiments. Section 4 discusses the theoretical implications, and Section 5 concludes.

2 Theoretical background

In this section, we will review the symmetric and asymmetric theories in detail, especially with respect to their predictions for the sentences we tested in our experiments. For concreteness, we will only discuss one symmetric approach, the theory of Križ 2015 and Križ 2016 and one asymmetric approach, the presuppositional implicature theory of Guerrini & Wehbe 2025, building on Bar-Lev 2018, Magri 2014 and Bar-Lev 2021. From now on, when we refer to the symmetric and asymmetric approaches, we sometimes refer to groups of theories, and sometimes to particular theories. It should be clear from the context of the discussion which is which.² We now review the two theories and their predictions in more detail.

2.1 The symmetric approach

Križ (2015, 2016) argues that homogeneity is a property of plural predication. A homogeneous predicate that holds of some but not all members of a plurality is undefined for that plurality.³ This has the effect of creating a gap between truth and falsity conditions and capturing the homogeneity pattern of sentences with plural definites. In the case of a simple positive sentence like (1a), the effect is that the sentence receives the third truth value if Frank opened some but not all of his presents.

$$(7) \quad \llbracket \text{Frank opened his presents} \rrbracket^w = \begin{cases} \text{TRUE} & \text{if Frank opened all of his presents in } w \\ \text{FALSE} & \text{if Frank opened none of his presents in } w \\ \# & \text{otherwise} \end{cases}$$

Negation just flips the truth and falsity conditions, leaving the gap intact, so (2a) is also mapped to the third truth-value if Frank opened only some of his presents.

² Other theories include the theory based on families of interpretations (Križ & Spector 2021) and the theory based on cognitive bias (Sbardolini 2023), which are versions of the symmetric approach. We leave an exploration of how these accounts relate in detail to our experimental results for future work.

³ This is an informal shorthand, which focuses on distributive predicates. For a more general formulation, applicable to both distributive and collective predicates, see Križ (2016, 2019).

$$(8) \quad \llbracket \text{Frank didn't open his presents} \rrbracket^w = \begin{cases} \text{TRUE} & \text{if Frank opened none of his presents in } w \\ \text{FALSE} & \text{if Frank opened all of his presents in } w \\ \# & \text{otherwise} \end{cases}$$

In order to account for non-maximal readings, [Križ 2016](#) proposes that perceived truth-values can differ from the semantic truth-values in certain contexts. That is, when a sentence that has a (non-trivially) trivalent meaning is used against a context where it denotes $\#$ in some of the possible worlds, it can be judged as effectively true—or as ‘true enough’ in Križ’s terminology—if $\#$ is enough to achieve the immediate conversational goal.

By way of illustration, consider again the Prohibited context above, in which we are trying to answer the question of whether Frank opened any of his presents. In other words, we would like to know if the possible world we are in is one where Frank opened none of his presents or one where he opened some or all of his presents. In Križ’s theory, the sentence in (1a) can be used in a context in which Frank opened only some of his presents to inform the interlocutors that the true answer to the question under consideration is positive, and that the rule has been broken, meaning Frank opened at least one of the presents, despite the fact that the semantic truth-value the sentence denotes is $\#$. This is because, in the Prohibited context, we do not really care about the difference between the possible worlds in which (1a) denotes TRUE and those in which it denotes $\#$. More in detail, what is relevant in the context can be modelled as a partition of the epistemically live worlds, often called ‘question under discussion’ (QuD, cf. [Roberts 1996](#), among many others; [Križ 2015](#) actually speaks of *current issues*, which is closely related to but strictly speaking different from QuDs. For our purposes the differences are immaterial). In some cases, the partition associated with the QuD of the context lumps the worlds where the sentence receives $\#$ with those where it receives TRUE by the truth-conditions in (7). In such cases, the proposition can be judged as effectively true, i.e., ‘true enough’. This is the case in the Prohibited context for (1a) given its truth-conditions in (7) as the worlds where Frank opened some but not all his presents are lumped with those where he opened all of them.

Consider now the Required context above again. Here the QuD is whether Frank opened all of his presents. In this case, a sentence like (1a) is semantically $\#$ but cannot be judged as effectively true. This is so because now the partition of the QuD lumps together the possible worlds in which the sentence denotes $\#$ and those in which it denotes FALSE. That is, the QuD lumps the worlds where Frank opened some but not all of the presents with those where he opened none.

Importantly for us, the same holds for the negative case in (2a) but in the opposite direction: in both the Required and the Prohibited context, in the non-maximal

scenario, (2a) receives the semantic value #. This time, however, a judgement of ‘effectively true’ is predicted to be available in the Required but not in the Prohibited context. Given the truth-conditions in (8), the QuD in the former context lumps worlds where the sentence receives # with those where it receives TRUE. Recall that the worlds where Frank opened some but not all presents are in the same equivalence class as those where he opened none of them. For the Prohibited context a judgement of ‘effectively true’ is not predicted. Since the QuD here lumps worlds where Frank opened some but not all with those where he opened all, the #-worlds share an equivalence class with the FALSE worlds given (8). In this sense, the theory makes symmetric predictions for positive vs. negative sentences with respect to non-maximality.

In sum, under this theory, definite plurals give rise to a non-bivalent meaning, which feeds pragmatics in such a way that non-classical truth-values can be deemed effectively true and effectively false, depending on the interlocutors’ collective interests. It is important to note at this point that such a theory is not the only way of making symmetric predictions regarding non-maximality in positive and negative environments. The alternative- and supervaluationist theory by Križ & Spector 2021, for instance, makes parallel predictions for the cases considered here but does not rely on trivalency. We group the theories therefore together for present purposes.⁴

2.2 The asymmetric approach

As noted above, we focus on the asymmetric approach based on presuppositional implicature developed by Guerrini & Wehbe (2025), since it is the only existing framework that provides a unified and transparent account of all the environments examined here, including, in particular, non-monotonic contexts.⁵ For expository purposes, however, we begin by sketching the implicature-based analysis proposed by Bar-Lev (2018, 2021), on which Guerrini & Wehbe’s account is built. We then turn to a more detailed presentation of the presuppositional implicatures approach.

2.2.1 The implicature-based theory

The implicature-based theory of Bar-Lev (2018, 2021) derives homogeneity from a basic existential semantics for plural definites (see also Magri 2014 for a similar

⁴ Križ & Spector 2021 argue that the proposal by Križ 2016 is too permissive, and overgenerates judgements of ‘effectively true’ in certain contexts, while their theory is more restrictive. For our immediate concerns the differences between the two theories are immaterial.

⁵ On the challenges faced by implicature-based approaches in those contexts see Gotzner, Romoli & Santorio 2020, Bar-Lev 2018, Bar-Lev & Fox 2020; we are grateful to an anonymous reviewer for discussion on this point.

idea). To illustrate the core idea, consider a context in which the relevant presents are commonly known to be a , b , and c , and nothing else. In such a context, the literal interpretation of (1a) is analyzed as in (9): Frank opened at least one of a , b , and c . This meaning is clearly too weak to capture the intuitively salient interpretation of (1a). By contrast, its negation in (10) straightforwardly yields the observed interpretation of (2a): Frank didn't open any of a , b , and c .

$$(9) \quad \exists x \in \{a, b, c\} (\text{frank's.present}(x) \wedge \text{opened}(\text{frank}, x))$$

$$(10) \quad \neg \exists x \in \{a, b, c\} (\text{frank's.present}(x) \wedge \text{opened}(\text{frank}, x))$$

In order to derive the strengthened interpretation in the positive case, the implicature-based theory maintains that the basic existential meaning is enriched by a scalar implicature, yielding the maximal interpretation in (11).

$$(11) \quad \forall x \in \{a, b, c\} (\text{frank's.present}(x) \rightarrow \text{opened}(\text{frank}, x))$$

Crucially for present purposes, the theory connects non-maximal readings to a weakening of this implicature, in a way that will be detailed below. Since implicature-based strengthening applies only in positive environments, this derives the observed asymmetry between positive and negative contexts discussed above. As we will see, the presuppositional implicature approach preserves this asymmetry, although it derives it in a somewhat different manner.

2.2.2 The presuppositional twist

The presuppositional implicatures account retains most of the core ingredients of the implicature-based approach, while integrating them into a presuppositional theory of homogeneity.

To illustrate the theory in more concrete terms, we expand on some of the details of the implicature-based analysis that are adopted in the presuppositional implicature account. The framework rests on several key assumptions. First, [Guerrini & Wehbe 2025](#) adopts [Bar-Lev \(2021\)](#)'s proposal that plural predication involves an existential pluralization operator $\exists\text{-PL}$, which is associated with a domain of quantification D .⁶

⁶ The meaning of the existential pluralization operator is given in (i) (where $\text{PART}(x)$ is the set of atomic parts of x): there exist some atomic parts of the domain D that satisfy the predicate P . When combined with the predicate, this yields an existential claim such as (9). We assume that the plural definite moves from the object position in order to make semantic composition work, but nothing crucial hinges on this assumption, and alternative analytical options are available.

$$(i) \quad \llbracket \exists\text{-PL}_D \rrbracket = \lambda P_{\langle e,t \rangle} \lambda x_e \exists y \in D \cap \text{PART}(x) [P(y)]$$

Assuming that the relevant presents are $\{a, b, c\}$, the interpretation of (12) is the existential meaning given in (9).

$$(12) \quad \llbracket \text{Frank [his presents]}_1 [\exists\text{-PLD opened } t_1] \rrbracket = \\ \exists x \in \{a, b, c\} (\text{frank's.present}(x) \wedge \text{opened}(\text{frank}, x))$$

Second, following Bassi, Del Pinal & Sauerland (2021) and Del Pinal, Bassi & Sauerland (2024), the theory assumes that scalar implicatures are associated with a silent operator PEX, which strengthens the presuppositional component of a sentence. More precisely, PEX takes a sentence (the prejacent) and a set of contextually salient alternatives as arguments, and returns the prejacent as its assertive content, while adding to its presuppositional component the negations of those alternatives that are innocently excludable (IE). The notion of innocent exclusion is defined in (13), where C denotes the set of relevant alternatives.

In addition to conjoining the prejacent with the negations of its IE alternatives, PEX also introduces a homogeneity presupposition over a subset of other alternatives, namely those that are innocently includable (II). Specifically, it requires that these alternatives be either all true or all false. Innocently includable alternatives, defined in (14), are those that belong to every maximal subset of alternatives that can be jointly satisfied with the prejacent and the negations of all IE alternatives, as illustrated in (14).

$$(13) \quad \text{IE}(p, C) := \bigcap \left\{ C' \mid \begin{array}{l} C' \text{ is a maximal subset of } C \text{ such that for some } w, \\ p(w) = \text{TRUE and each member of } C' \text{ denotes FALSE at } w \end{array} \right\}$$

$$(14) \quad \text{II}(p, C) := \bigcap \left\{ C'' \mid \begin{array}{l} C'' \text{ is a maximal subset of } C \text{ such that for some } w, \\ p(w) = \text{TRUE} \\ \text{and each member of } \text{IE}(p, C) \text{ denotes FALSE at } w \\ \text{and each member of } C'' \text{ denotes TRUE at } w \end{array} \right\}$$

Taken together, these assumptions yield the definition of PEX in (15). The operator leaves the prejacent intact at the level of assertion and adds to its presuppositional component the negations of all innocently excludable alternatives, together with a homogeneity presupposition over the innocently includable alternatives.

$$(15) \quad \llbracket \text{PEX} \rrbracket (C)(p)(w) \\ \text{asserts: } p(w) \\ \text{presupposes:} \\ \forall \phi \in \text{IE}(p, C) [\neg \llbracket \phi \rrbracket (w)] \\ \forall \phi \in \text{II}(p, C) [\llbracket \phi \rrbracket (w)] \vee \neg \exists \phi \in \text{II}(p, C) [\llbracket \phi \rrbracket (w)]$$

With these definitions in place, we can show that the strengthened interpretation of (1a) corresponds to the maximal universal reading in (11). This follows from the assumption that the alternatives to (12) are obtained by associating $\exists$ -PL with proper subdomains of D , as in (16).

$$(16) \quad \left\{ \begin{array}{l} \text{Frank [his presents]}_1 [\exists\text{-PL}_{\{a,b,c\}} \text{ opened } t_1], \\ \text{Frank [his presents]}_1 [\exists\text{-PL}_{\{a,b\}} \text{ opened } t_1], \\ \text{Frank [his presents]}_1 [\exists\text{-PL}_{\{b,c\}} \text{ opened } t_1], \\ \text{Frank [his presents]}_1 [\exists\text{-PL}_{\{a,c\}} \text{ opened } t_1], \\ \text{Frank [his presents]}_1 [\exists\text{-PL}_{\{a\}} \text{ opened } t_1], \\ \text{Frank [his presents]}_1 [\exists\text{-PL}_{\{b\}} \text{ opened } t_1], \\ \text{Frank [his presents]}_1 [\exists\text{-PL}_{\{c\}} \text{ opened } t_1] \end{array} \right\}$$

None of the alternatives in (16) is innocently excludable, whereas all of them are innocently includable. Therefore a homogeneity presupposition about all of them is added, which can be stated as in (17): either Frank opened all of his presents or he opened none of them.

$$(17) \quad \begin{array}{l} \llbracket \text{PEX}[\text{Frank [his presents]}_1 [\exists\text{-PL}_D \text{ opened } t_1]] \rrbracket \\ \text{asserts: } \exists x \in \{a, b, c\} (\text{frank's.present}(x) \wedge \text{opened}(\text{frank}, x)) \\ \text{presupposes: } \forall x \in \{a, b, c\} (\text{frank's.present}(x) \rightarrow \text{opened}(\text{frank}, x)) \vee \neg \exists x \in \{a, b, c\} (\text{frank's.present}(x) \wedge \text{opened}(\text{frank}, x)) \end{array}$$

At this point, the account may appear to amount to a more complicated variant of the symmetric approach, differing only in the source of homogeneity. This impression is partly correct. However, there are a few important differences. In particular, the crucial difference for us is that they analyze definedness conditions as presuppositions. While the motivation for this move is not directly relevant here, it is important that this treatment allows for presupposition accommodation, rather than reliance on the pragmatic re-interpretation of undefinedness proposed by Križ (2015, 2016). This difference underlies the divergent predictions of the asymmetric approach with respect to non-maximality as we will see.

That is, it is assumed that presuppositions are globally accommodated by an operator A , which effectively conjoins presuppositional and assertive content. As a result, the presupposition and the assertion are collapsed into a single truth-conditional meaning. In other words, the basic existential meaning of sentences such as (1a) is strengthened to a universal interpretation via a maximality presuppositional implicature, which is subsequently integrated into the asserted content. The resulting meaning in (18) is the intuitively correct maximal reading conveying that Frank opened all of his presents.

$$(18) \quad \begin{aligned} & \llbracket A[\text{PEX}[\text{Frank} [\text{his presents}]_1 [\exists\text{-PL}_D \text{ opened } t_1]]] \rrbracket \\ & = \forall x \in \{a, b, c\} (\text{frank's.present}(x) \rightarrow \text{opened}(\text{frank}, x)) \end{aligned}$$

As we said, in negative environments, the presuppositional implicature approach retains the asymmetric predictions of the implicature-based theory. To see why, consider (2a). Following [Guerrini & Wehbe 2025](#), suppose that PEX applies within the scope of negation, thereby introducing the same homogeneity presupposition, which then projects globally, suggesting that Frank opened all or none of his presents.⁷

$$(19) \quad \begin{aligned} & \llbracket \text{not}[\text{PEX}[\text{Frank} [\text{his presents}]_1 [\exists\text{-PL}_D \text{ opened } t_1]]] \rrbracket \\ & \text{asserts: } \neg \exists x \in \{a, b, c\} (\text{frank's.present}(x) \wedge \text{opened}(\text{frank}, x)) \\ & \text{presupposes: } \forall x \in \{a, b, c\} (\text{frank's.present}(x) \rightarrow \text{opened}(\text{frank}, x)) \vee \neg \exists x \in \\ & \quad \{a, b, c\} (\text{frank's.present}(x) \wedge \text{opened}(\text{frank}, x)) \end{aligned}$$

In this configuration, however, unlike in the positive case, the assertive component entails the presupposition. Consequently, after accommodation, the strengthened meaning is equivalent to the assertive component: Frank opened none of his presents. As we will see, this is crucial in predicting the asymmetry of non-maximality.

$$(20) \quad \begin{aligned} & \llbracket A[\text{not}[\text{PEX}[\text{Frank} [\text{his presents}]_1 [\exists\text{-PL}_D \text{ opened } t_1]]] \rrbracket = \\ & \neg \exists x \in \{a, b, c\} (\text{frank's.present}(x) \wedge \text{opened}(\text{frank}, x)) \end{aligned}$$

Finally, in this theory, as in the implicature-based approach, non-maximal readings arise as a result of contextual modulation of the maximality presuppositional implicature. Specifically, while the strengthened interpretation of the positive example above is obtained by quantifying over the full set of alternatives in (16), in certain contexts some of these alternatives may be ignored, or “pruned.” Such pruned sets of alternatives yield weaker presuppositions, which in turn give rise to non-maximal readings.

For instance, strengthening (1a) relative to the alternatives in (21)—a set from which the atomic alternatives have been pruned—yields the interpretation in (22), which can be paraphrased as: Frank opened at least two of *a*, *b*, and *c*.

$$(21) \quad \left\{ \begin{array}{l} \text{Frank} [\text{his presents}]_1 [\exists\text{-PL}_{\{a,b,c\}} \text{ opened } t_1], \\ \text{Frank} [\text{his presents}]_1 [\exists\text{-PL}_{\{a,b\}} \text{ opened } t_1], \\ \text{Frank} [\text{his presents}]_1 [\exists\text{-PL}_{\{b,c\}} \text{ opened } t_1], \\ \text{Frank} [\text{his presents}]_1 [\exists\text{-PL}_{\{a,c\}} \text{ opened } t_1] \end{array} \right\}$$

⁷ They in fact assume that PEX applies at every scope site. We abstract away from this complication here, because, as far as we can see, it does not affect the present discussion here and for all cases we tested. Therefore, we only consider LFs where PEX is added at the lowest scope site.

$$(22) \quad \llbracket A[\text{PEX}[\text{Frank} [\text{his presents}]_1 [\exists\text{-PL}_D \text{ opened } t_1]]] \rrbracket = \\ \exists x \in \{a, b\} (\text{frank}'s.\text{present}(x) \wedge \text{opened}(\text{frank}, x)) \wedge \\ \exists x \in \{b, c\} (\text{frank}'s.\text{present}(x) \wedge \text{opened}(\text{frank}, x)) \wedge \\ \exists x \in \{a, c\} (\text{frank}'s.\text{present}(x) \wedge \text{opened}(\text{frank}, x))$$

While the modulation of the maximality presuppositional implicature via pruning is in principle available in both positive and negative environments, the fact that, in negative contexts, the assertive component entails the homogeneity presupposition renders this modulation truth-conditionally inert. That is, given that the prejacent already entails the homogeneity presupposition, weakening the latter further has no effect on the overall interpretation.

As a consequence, non-maximal readings are not predicted under negation. This yields the observed polarity asymmetry, in contrast to the symmetric approach discussed above.

In this section, we have illustrated the predictions of the two theories for simple positive and negative sentences. For the more complex structures investigated in our experiments, the same general pattern obtains, modulo additional complications in non-monotonic environments. For reasons of readability, we defer a detailed discussion of these predictions to the presentation of the individual experiments below.

3 The experiments

3.1 Overview of the design

We report on three web-based experiments that use a truth-value judgment task with a five-point Likert scale, building on [Augurzky et al. 2023](#). The design of the experiments is largely based on [Križ & Chemla 2015](#). As we are interested in the interpretation of definite plurals in environments with different polarity profiles, it is important to make sure that they do not outscope the operator that is responsible for the polarity of the local environment for the definite plural in question. As mentioned above, Križ & Chemla used bound pronouns for this purpose, and we followed suit.

We designed our target sentences as follows. They all contain a definite plural with a pronoun bound by one of the following four quantifiers, which have different polarity profiles with respect to their nuclear scope: *every*, *no*, *not every*, and *exactly two*. The actual sentences we tested looked as follows. The gender of the children was counterbalanced between participants in each experiment.

- (23) a. Every boy/girl opened his/her presents.
b. No boy/girl opened his/her presents.
c. Not every boy/girl opened his/her presents.

- d. Exactly two boys/girls opened their presents.

Experiment 1 tested (23a) and (23b), and Experiment 2 tested (23a) and (23c). Experiment 3 tested different aspects of the meaning of (23d), as will be explained below.

As will be explained in detail later, each experiment tested the linguistic stimuli in different conversational contexts, which varied between participants. Since we expected holding contextual information in memory to be cognitively demanding if contexts changed during the course of an experiment, we decided to test one and only one context per participant, which was introduced at the beginning of the experiment. Also, since each context is highly specific and mentions a particular family's rules for opening presents, we only tested the sentences in (23).

We started from (23a) and (23b) to first establish whether the previously found asymmetry would emerge also when controlling for contextual relevance. We then extended the investigation to compare the positive case to another negative environment, (23c), a comparison which, not in our intuitions did not exhibit the same asymmetry with (23a) (i.e., it seemed to allow non-maximality to the same extent).⁸ After confirming the intuition experimentally, we moved to the (23d) case, because of its properties of being both 'positive' and 'negative' at the same time, which allowed us to test the contrast within the same environment, so to speak.

One crucial difference from Križ & Chemla 2015 is that these sentences were evaluated against an explicit background context. We prepared two types of background context, similarly to the Required and Prohibited context described above (just adapted to more than one character; see below). Contexts varied between participants in each experiment. The idea, as above, was that the Required context would make salient the question whether or not all the presents are open, while the Prohibited context would make salient the question whether any present was open at all.

Each participant was randomly assigned to one of the contexts and at the beginning of the experiment, introduced to a family with four children (either all boys or all girls) with pictures for the characters,⁹ and told that each year, they give presents to each other at three different events: Christmas, Easter, and what they call the "Family Day". They celebrate these events with the children's grandparents, who usually arrive at noon. Then, the family rule was stated, which was different for the two contexts.

⁸ We were further motivated by the results in Geurts 2002 on donkey sentences, which show symmetric behaviour between *every* and *not every*. To the extent that this parallels our intuitions about plural definites, it provides an additional reason to examine the comparison more closely.

⁹ The pictures were constructed based on 12 free images downloaded from <https://www.irasutoya.com/>. The copyright of the images belongs to the designer.

- In the **Prohibited** context, it is explained that the family opens the presents together with the children’s grandparents, who come from a different city, so the children have to wait until they arrive before opening the presents they received. Then the family rule is stated on the next screen as “Opening the presents is prohibited before the guests arrive”. (Implicit QUD: Did any boy open any of his presents?)
- In the **Required** context, it is explained that the family celebrates the events together with the children’s aunt and uncle, who live close by, and the children open the presents early in the morning before their aunt and uncle arrive, so that the apartment is in an orderly state by then. Then on the next screen the rule is stated as “Opening the presents is required before the guests arrive”. (Implicit QUD: Did every boy open all of his presents?)

We made sure that all the descriptions about the rules contained no quantifiers, so as to avoid encouraging participants to contrast the interpretations of the target definite plural *his/her presents* and quantified phrases like *all his/her presents* (the use of a definite in the context rule seemed to us relatively innocuous in this respect). Part of the instructions mentioned that the family rule is sometimes not obeyed.

Each trial of the three experiments consisted of a picture, depicting what happened in one of the events and the year in which it happened (varying from 2015 to 2019). Participants performed two tasks. The first task asked them to judge if the family rule was obeyed in the event depicted by the picture, by choosing between two radio buttons that were labeled “yes” and “no”. This was done to make sure that participants understood and remembered the conversational context throughout the experiment. To remind them of the rule, the rule was displayed again on top of the screen in the same phrasing as in the instructions. The second task was to judge how well the sentence describes the scenario depicted in the picture on a five-point Likert scale ranging from ‘completely true’ to ‘completely false’. Each trial was designed as a single web page, whose response area looked as in Figure 1.

3.2 Predictions

We now return to the predictions of the two theories with respect to the contextual manipulations described above and the sentences tested in the experiments reported below.

3.2.1 The symmetric approach

We have illustrated the predictions of the symmetric approach for simple positive and negative sentences. For more complex sentences such as those tested in our

Were the rules respected?

☐

yes

☐

no

Kim opened her presents.

completely true

☐
☐
☐
☐
☐

completely false

Next

Figure 1 Response buttons used in the three experiments, with an example (filler) sentence “Kim opened her presents”. A picture depicting who opened which presents is placed right above this, which in turn was placed below the rule statement accompanied by pictures of the childrens’ parents.

experiments, repeated in (24), an explicit account of the projection of homogeneity is required. In order to save space, we do not discuss the available theoretical options for projection in detail here, and instead refer the reader to [Križ 2015](#) for extensive discussion. We focus on one of the most standard choices, namely Strong Kleene, and illustrate its predictions for the target scenarios described below. For present purposes, the differences between the Strong Kleene algorithm and the system proposed by [Križ’s \(2015\)](#) are immaterial.

- (24)
- a. Every boy opened his presents.
 - b. No boy opened his presents.
 - c. Not every boy opened his presents.
 - d. Exactly two boys opened their presents.

According to the Strong Kleene projection algorithm, the third truth value # is treated in a supervaluational manner. Roughly speaking, all possible ways of mapping # to either TRUE or FALSE are considered. If these different resolutions yield the same overall truth value for the sentence, the sentence is assigned that value; otherwise, it is assigned #.

A maybe clearer way of characterizing the resulting definedness conditions for the sentences in (24) is by paraphrasing the conditions under which they are either true or false. In particular, sentence (24a) is defined if either every boy opened all of

his presents, in which case it is true, or at least one boy opened none of his presents, in which case it is false. Parallel conditions apply to (24b) and (24c). Sentence (24d) is slightly more complex: it is defined if either exactly two boys opened all of their presents and no other boy opened any of his presents, or more than two boys opened all of their presents, or fewer than two boys opened any of their presents. The relevant conditions are summarized in (25).

- (25)
- a. Either every boy opened all of his presents or at least one boy opened none of his presents.
 - b. Either every boy opened none of his presents or at least one boy opened all of his presents.
 - c. Either at least one boy opened none of his presents or every boy opened all of his presents.
 - d. Either two boys opened all of their presents and no other boy opened any of his presents, or more than two boys opened all of their presents, or fewer than two boys opened any of their presents.

Given these conditions, it is straightforward to derive the predictions for the target scenarios used in our experiments. Consider (24a) in the top left scenario in Figure 2. In this scenario, (24a) denotes #, since the condition in (25a) is not satisfied. The same holds for (24b) and (24c) in the top right and top left scenarios, respectively. Finally, (24d) is undefined in both bottom scenarios, as (25d) is not satisfied in either case.

In sum, the underlying trivalent semantics predicts all sentences in (24) to denote # in the target scenarios. As in the simpler cases discussed above, however, Kriz's theory allows for non-maximality in all of these cases via pragmatic reinterpretation of #, depending on contextual relevance. For example, in the prohibited context where intuitively the question under discussion concerns whether any presents was open at all, i.e., whether any boy opened any of his presents, (24a) is predicted to be 'true enough' for the same reason as in the simple positive case (the partition of the context is such that the worlds in which the sentence is undefined are in the same cell as those in which it is true; worlds in which every boy opened all of his presents). Therefore the expectation is that a sentence like (24a) will be accepted more in such a context than in the required context, where the current issue is whether every boy opened all of his presents. In this latter case, like in the simple positive case, the worlds in which the sentence is undefined will be grouped together with the ones in which the sentence is false.

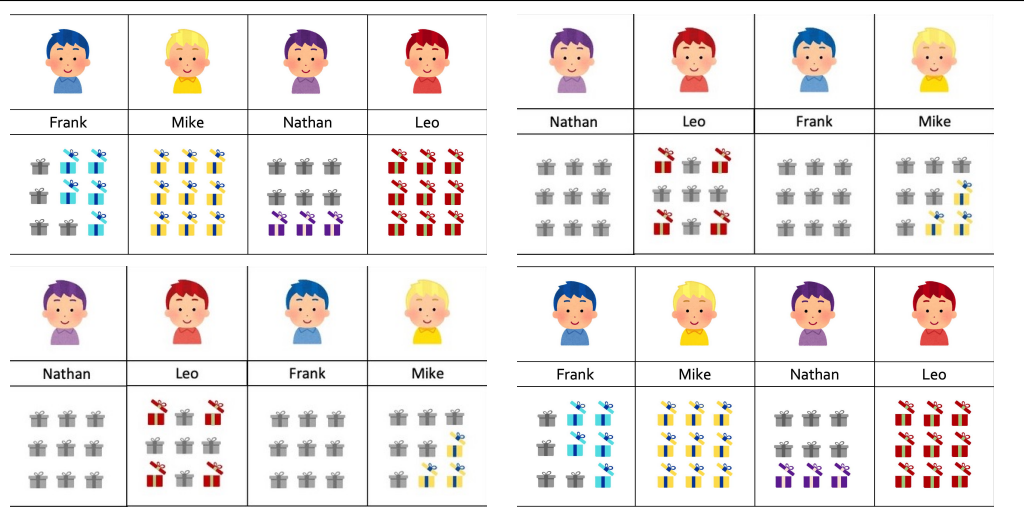


Figure 2 Examples of visual scenarios used in Experiments 1–3. The top left picture is the target scenario for *every* and *not every*, the top right picture for *no*. The bottom pictures are the target scenarios for *exactly two*.

3.2.2 The asymmetric approach

Like the symmetric approach, the presuppositional implicature framework must account for the projection of homogeneity in complex environments. [Guerrini & Webbe 2025](#) explicitly adopt the Strong Kleene projection principle discussed above and therefore make the same predictions concerning projection. As mentioned, however, a crucial difference is that they analyze definedness conditions as presuppositions, which in turn allows for presupposition accommodation, rather than reliance on the pragmatic re-interpretation of [Križ \(2015\)](#). This difference underlies the divergent predictions of the asymmetric approach with respect to non-maximality.

We illustrate this by considering the four cases in turn. Consider first (24b) and (24c), and assume that PEX is inserted in the scope of the quantifiers, as in (26) and (27). These are negative environments. As in the simple negative cases discussed above, the assertive component entails the homogeneity presupposition (cf. (25b) and (25c)). Consequently, once the presupposition is accommodated, the resulting meaning coincides with the assertive component: ‘no boy opened any of his presents’ and ‘not every boy opened any of his presents’, respectively.

$$(26) \quad \llbracket A[\text{[no boy]}_2 \text{ PEX}[t_2 \text{ [his presents]}_1 [\exists\text{-PL}_D \text{ opened } t_1]]] \rrbracket = \\ \neg \exists x[\text{boy}(x) \wedge \exists y \in \text{present.of}(y, x)[\text{opened}(x, y)]]$$

$$(27) \quad \llbracket A[\text{[not every boy]}_2 \text{ PEX}[t_2 \text{ [his presents]}_1 [\exists\text{-PL}_D \text{ opened } t_1]]] \rrbracket = \\ \neg \forall x[\text{boy}(x) \rightarrow \exists y \in \text{present.of}(y, x)[\text{opened}(x, y)]]$$

As a result, weakening the presuppositional implicature through pruning, although in principle possible, has no effect on the overall truth conditions of these sentences. Non-maximality is therefore predicted to be unavailable in these cases.

The situation is different for (24a). Since (24a) occurs in a positive environment, the assertive component is weaker than the homogeneity presupposition. After accommodation, the resulting meaning can be paraphrased as ‘every boy opened all of his presents’. Moreover, as discussed above, the presuppositional implicature may be contextually modulated, making (24a) assertable in non-maximal scenarios.

$$(28) \quad \llbracket A[\llbracket \text{every boy} \rrbracket_2 \text{ PEX}[t_2 \llbracket \text{his presents} \rrbracket_1 \llbracket \exists\text{-PLD opened } t_1 \rrbracket]] \rrbracket = \\ \forall x[\text{boy}(x) \rightarrow \forall y \in \text{present.of}(y, x)[\text{opened}(x, y)]]$$

Thus, the presuppositional implicature approach predicts an asymmetry between (24a), on the one hand, and (24b) and (24c), on the other, with respect to non-maximality.

The predictions for (24d) are more complex. Following standard practice, we decompose non-monotonic quantifiers into positive and negative components. The assertion and presupposition may be paraphrased as in (29).

- (29) **ASSERTION:** Two boys opened at least one of their presents and no other boy opened any of his presents.
 PRESUPPOSITION: Two boys opened all of their presents and no other boy opened any of his presents (or more than two boys opened all of their presents, or fewer than two boys opened any of their presents)

Informally, the positive component of the assertion is weaker than its presuppositional counterpart, whereas the negative component of the assertion entails its presuppositional counterpart. When assertion and presupposition are combined, the positive component is strengthened, while the negative component remains unchanged. The resulting meaning can be paraphrased as ‘two boys opened all of their presents and no other boy opened any of his presents’.

$$(30) \quad \llbracket A[\llbracket \text{exactly two boys} \rrbracket_2 \text{ PEX}[t_2 \llbracket \text{their presents} \rrbracket_1 \llbracket \exists\text{-PLD opened } t_1 \rrbracket]] \rrbracket = \\ \exists X[|X| = 2 \wedge \forall y \in \text{PART}(X) \rightarrow \text{boy}(y) \wedge \forall z \in \text{present.of.}(z, y)[\text{opened}(y, z)]] \wedge \\ \forall t \in \text{boy}[\neg \text{PART}(X) \rightarrow \neg \exists z \in \text{presents.of}(z, t)[\text{opened}(t, z)]]$$

This is crucial for present purposes, since it implies that weakening the presupposition can affect the positive component of the overall interpretation, but not the negative one. Non-maximality is therefore predicted to be available only for the positive component of non-monotonic sentences. An internal asymmetry is thus predicted within such constructions.

SENTENCE	CONTEXT	
	PROHIBITED	REQUIRED
Every boy opened his presents	✓	×
No boy opened his presents	×	☒
Not every boy opened his presents	×	☒
Exactly two boys opened their presents (<i>positive part</i>)	✓	×
Exactly two boys opened their presents (<i>negative part</i>)	×	☒

Table 1 Predictions of the symmetric approach for the tested sentences with respect to non-maximality and context manipulation.

SENTENCE	CONTEXT	
	PROHIBITED	REQUIRED
Every boy opened his presents	✓	×
No boy opened his presents	×	☐
Not every boy opened his presents	×	☐
Exactly two boys opened their presents (<i>positive part</i>)	✓	×
Exactly two boys opened their presents (<i>negative part</i>)	×	☐

Table 2 Predictions of the asymmetric approach for the tested sentences with respect to non-maximality and context manipulation.

In sum, the asymmetric approach differs from the symmetric one in predicting a restricted distribution of non-maximal readings. While non-maximality is expected in non-monotonic cases, it is predicted to arise only in the positive component of such sentences.

The predictions of the symmetric approach are summarized in Table 1, and those of the asymmetric approach in Table 2. The symmetric approach predicts non-maximality for all tested sentences, with comparable effects of context manipulation in positive and negative environments, albeit in opposite directions. By contrast, the asymmetric approach predicts non-maximality and context effects only in positive environments, including the positive component of non-monotonic cases.

3.3 Experiment 1

3.3.1 Methods

In our first experiment, we compared sentences involving the quantifiers *every* and *no* with a plural definite in their scope:

- | | | | |
|------|----|---|----------|
| (31) | a. | Every boy/girl opened his/her presents. | POSITIVE |
| | b. | No boy/girl opened his/her presents. | NEGATIVE |

Each sentence was presented in the context of one of three types of pictures. Each picture showed four children, each with nine presents, as in the examples in Figure 3. The pictures in the CONTROL conditions (e.g., the top two pictures in Figure 3) depicted uniform situations, either with all the presents closed (which were greyed out in the pictures), or with all the presents open. The sentences should receive straightforward true or false responses in these conditions. In particular, the control pictures that should elicit true responses for sentences with *every* should elicit false responses for sentences with *no* and vice versa.

The pictures in the TARGET condition depict non-maximal scenarios, where two of the four children opened some but not all of their presents. We prepared two types of TARGET pictures that differ with respect to what the other two children did, namely, pictures where the other children opened all of their presents (e.g., the bottom left picture in Figure 3) and pictures where they opened none of their presents (e.g., the bottom right picture in Figure 3). We tested both sentences, with *every* and with *no*, against both types of target pictures. Given that their non-maximal readings are true in different target pictures, however, only data from those pictures that could make the sentences true under a non-maximal reading were included in the analysis (i.e., for sentences with *every*, pictures similar to the bottom left one in Figure 3; for sentences with *no*, pictures similar to the bottom right one in Figure 3).

As explained above, each participant was assigned either the PROHIBITED context or the REQUIRED context, and after reading the instructions did one practice item. The practice item displayed the same four children, one of whom opened all of their presents, one of whom opened none of their presents, and the other two of whom opened some but not all of their presents. The sentence for the practice item was a non-quantified sentence mentioning one of the children by name, e.g., “Leo opened his presents”. No feedback was given.

The experimental factors manipulated were CONTEXT (PROHIBITED or REQUIRED), PICTURE (TARGET, CONTROL-TRUE, CONTROL-FALSE) and POLARITY (POSITIVE = *every* or NEGATIVE = *no*), so $2 \times 3 \times 2 = 12$ conditions in total. Among these, CONTEXT was a between-participant factor, and the other two were within-participant. We counterbalanced the position of the *yes* and *no* buttons for the first

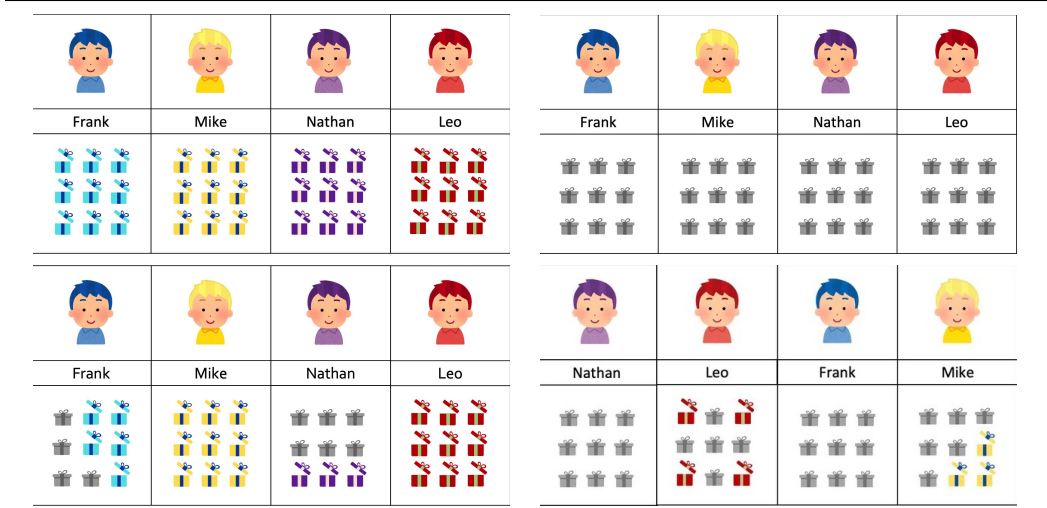


Figure 3 Examples of pictures used in Experiment 1. The top two pictures are items in the CONTROL conditions. The bottom left picture was a target item for *every* and the bottom right picture was a target item for *no*.

task and the order of the end points of the Likert scale, such that the ‘yes’ button was always on the same side as ‘Completely true’, as well as the gender of the children (boys or girls).

There were therefore 8 lists ($2 \text{ contexts} \times 2 \text{ genders} \times 2 \text{ ways of labeling the response buttons}$). Each list contained 24 experimental items, half with the positive quantifier *every* and the other half with the negative quantifier *no*. For the two tested sentences, with *every* and with *no*, there were 4 CONTROL-TRUE items, 4 CONTROL-FALSE items, and 4 TARGET items. In other words, each list contained 24 experimental items, half with the positive quantifier *every* and the other half with the negative quantifier *no*. As mentioned above, 2 of the TARGET items would not make the sentence true regardless of non-maximality, so we excluded them from the analysis. Consequently, for each participant, we analysed the data of 2 items for each Target condition. The experimental items appeared in a randomized order for each participant.

192 native speakers of English were recruited on Prolific (www.prolific.co) and paid £1.50 for their participation. 4 participants were excluded for low accuracy (75% or less correct) on the main judgment task on the CONTROL items, where responses above 3 for CONTROL-FALSE and responses below 3 for CONTROL-TRUE were considered to be errors. For the remaining participants, the mean accuracy of the CONTROL conditions was 98.1%.

SENTENCE	CONTEXT	
	PROHIBITED	REQUIRED
Every boy opened his presents	✓	×
No boy opened his presents	×	☒

Table 3 Predictions of the symmetric approach.

SENTENCE	CONTEXT	
	PROHIBITED	REQUIRED
Every boy opened his presents	✓	×
No boy opened his presents	×	☐

Table 4 Predictions of the asymmetric approach.

3.3.2 Predictions

Remember that the asymmetric approach predicts an effect of context on the availability of the non-maximal readings only in the case of *every* but not in the case of *no*. Conversely, the symmetric approach predicts a similar effect of context in the two cases, albeit going in opposite directions. Tables 3 and 4 summarize the relevant predictions again.

3.3.3 Results

Figure 4 shows the mean ratings for the experimental conditions. We have recoded CONTEXT into LAX (PROHIBITED for *every*, REQUIRED for *no*) and STRICT (REQUIRED for *every*, PROHIBITED for *no*), because the two contexts should have different effects on the two sentences. LAX contexts should improve the rating in the TARGET condition (because they should make non-maximal readings easier to access), while STRICT contexts should not (because they should make non-maximal readings harder to access).

We fitted a proportional-odds cumulative logit mixed effects model to the data from the TARGET condition, using `clmm()` of the `ordinal` package (Christensen 2022) on R (R Core Team 2022).¹⁰ CONTEXT was sum-coded (STRICT = −0.5; LAX = 0.5), while POLARITY was treatment-coded with *no* as the reference level

¹⁰ While we would have ideally included the CONTROL conditions in the model, the floor and ceiling effects led to convergence issues. However, since performance on the CONTROL conditions is near-perfect, very little information is lost by excluding them.

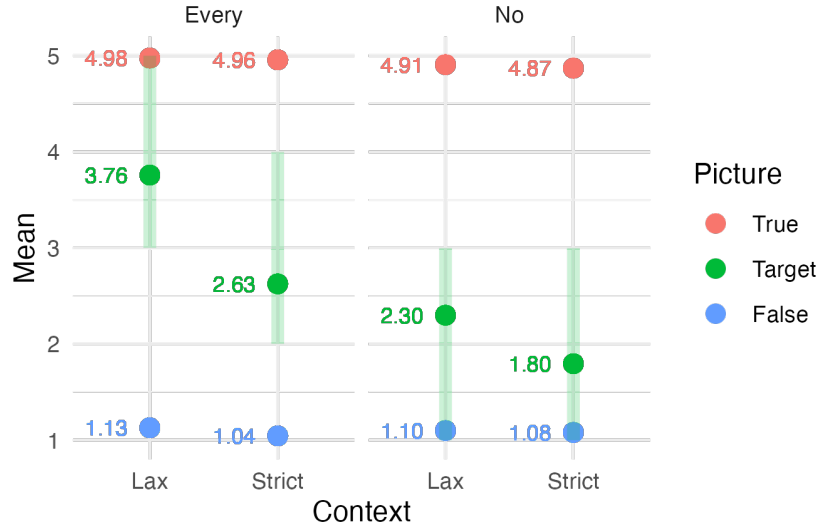


Figure 4 Mean ratings in Experiment 1. Bars indicate interquartile range. CONTEXT is recoded as LAX (favors non-maximality) or STRICT (favors maximality).

(this choice was dictated by wanting to directly test the effect of context on 'no', as that is where there is theoretical disagreement). These and their interaction were the fixed effects of the model. The random effect structure was maximal in the sense of Barr et al. 2013: by-participant variation for the intercept and POLARITY, and their correlation, as well as by-item variation for the intercept.¹¹

The model estimates are summarised in Table 5. The p -values were obtained via likelihood ratio tests with simpler models using `anova()` on R. The model revealed a significant simple effect of CONTEXT (i.e., CONTEXT had an effect on *no*; $\chi^2(1) = 12.7$; $p < 0.001$) and a significant main effect of POLARITY (*every* is more acceptable than *no*; $\chi^2(1) = 52.2$; $p < 0.001$), as well as a significant interaction (*every* is more sensitive to CONTEXT than *no*; $\chi^2(1) = 10.9$; $p < 0.001$).

3.3.4 Discussion

The main findings of Experiment 1 are the following:

- i. Definite plurals under *every* received higher ratings than definite plurals under *no* in non-maximal scenarios.

¹¹ As shown in Table 5, the by-item random intercept term does not explain any variance. Leaving it out makes no difference in the estimates of the other coefficients.

Fixed effects	β	s.e.	p
θ_1	-0.49	0.21	
θ_2	1.34	0.23	
θ_3	3.07	0.28	
θ_4	7.23	0.51	
CONTEXT	1.41	0.40	< 0.001***
POLARITY	3.36	0.37	< 0.001***
CONTEXT $\times$ POLARITY	2.43	0.74	< 0.001***
Random effects	σ^2	σ	ρ
By-participant intercept	4.62	2.15	
By-participant POLARITY	13.36	3.65	-0.42
By-item intercept	< 0.001	< 0.001	

Table 5 Estimates of the cumulative logit mixed effects model for Experiment 1. The p -values were obtained via likelihood ratio tests.

- ii. CONTEXT had an effect on the rating of definite plurals under *no* in non-maximal scenarios.
- iii. CONTEXT had a larger effect on definite plurals under *every* than those under *no*.

Recall that according to the asymmetric approach, definite plurals may receive non-maximal readings under *every* but not under *no*, so it predicts CONTEXT to have an effect in the former case but not in the latter. Under the symmetric approach, definite plurals may receive non-maximality readings under both *every* and *no*, so it predicts parallel effects of CONTEXT.

The first and last finding above are therefore in line with the predictions of the asymmetric approach, while the second one is unexpected. Conversely, the second finding is in line with the expectations of the symmetric approach, while the other two pose challenges.

3.4 Experiment 2

3.4.1 Methods

The design of Experiment 2 is essentially parallel to that of Experiment 1, with the negative quantifier being replaced by *not every*. This quantifier is an interesting case because it also creates a negative grammatical environment for the plural definite, so

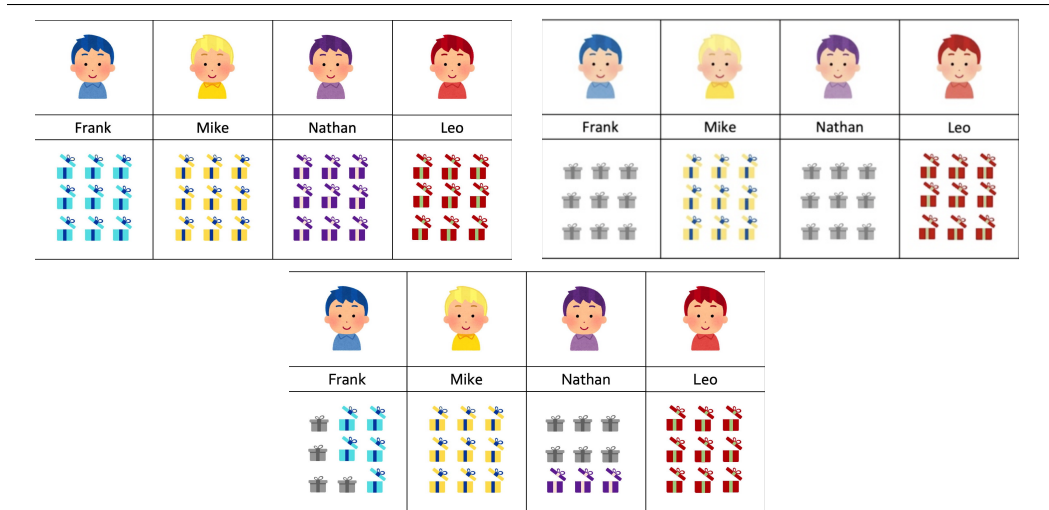


Figure 5 Examples of pictures used in Experiment 2. The top two pictures are items in the CONTROL condition and the bottom picture is an item in the TARGET condition.

the predictions of the two approaches diverge here in the same way as in the case of *no*, yet our intuitions were that it would behave differently from *no*.

- (32) a. Every boy/girl opened his/her presents. POSITIVE
b. Not every boy/girl opened his/her presents. NEGATIVE

One immediate consequence of this change is that the same TARGET pictures could be used for both sentences. That is, for all the items in the TARGET condition, two children opened some but not all of their presents, while the other two opened all of their presents. The CONTROL and TARGET pictures are illustrated in Figure 5.

The experimental sentences were spread over 8 lists (2 contexts $\times$ 2 genders $\times$ 2 ways of labeling the response buttons). Each list contained 32 items as a whole, with 24 of them being experimental and 8 of them being filler items. Half of the experimental items contained the positive quantifier *every* and the other half the negative quantifier *not every*. For each experimental sentence, there were 4 CONTROL-TRUE items, 4 CONTROL-FALSE items, and 4 TARGET items. Filler items were included in this experiment, because in the PROHIBITED condition, the correct answer to the *yes/no* task was uniformly *no* for all the items in the TARGET and CONTROL conditions. The filler items consisted of four pictures in which none of the children opened any of their presents, so that the correct answer for the *yes/no* task was *yes* in the PROHIBITED condition. Pictures were accompanied by sentences

SENTENCE	CONTEXT	
	PROHIBITED	REQUIRED
Every boy opened his presents	✓	×
Not every boy opened his presents	×	☒

Table 6 Predictions of the symmetric approach.

SENTENCE	CONTEXT	
	PROHIBITED	REQUIRED
Every boy opened his presents	✓	×
Not every boy opened his presents	×	☐

Table 7 Predictions of the asymmetric approach.

containing proper names in place of the quantifiers. 4 of these sentences were positive and 4 of them were negative as in (33).

- (33) a. Frank opened his presents.
b. Frank didn't open his presents.

The filler items were not used in the analysis.

192 native speakers of English were recruited from Prolific (www.prolific.co) and paid £1.50 for their participation. Six were excluded for low (75% or less) accuracy on the CONTROL items. The remaining participants' accuracy rate on the CONTROL items was 97.4%.

3.4.2 Predictions

Recall that the asymmetric approach predicts an effect of context on the availability of the non-maximal readings only in the case of *every* but not in the case of *not every*. Conversely, the symmetric approach predicts a similar effect of context in the two cases again. Tables 6 and 7 summarize the relevant predictions again.

3.4.3 Results

Figure 6 shows the mean ratings of Experiment 2. As it indicates, some of the control conditions had less-than-perfect accuracy. In particular, *every* in the FALSE CONTROL condition received a relatively high rating in comparison to the cor-

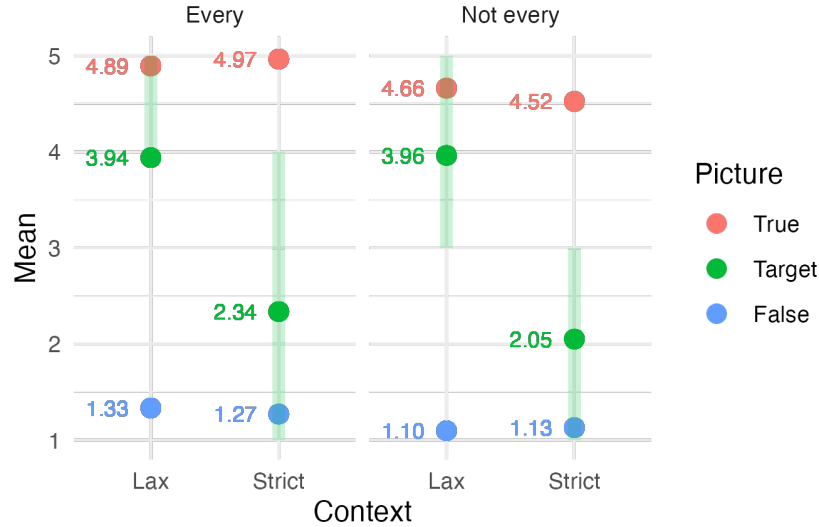


Figure 6 Mean ratings in Experiment 2. Bars indicate interquartile range. CONTEXT is recoded as LAX (favors non-maximality) or STRICT (favors maximality).

responding condition in Experiment 1. Importantly, we checked if there is any indication that the performance in the CONTROL conditions was consequential by fitting a proportional-odds cumulative logit mixed effects model to the data from the CONTROL conditions. The model had CONTEXT, POLARITY, and PICTURE (CONTROL-TRUE vs. CONTROL-FALSE) and all their interactions as fixed effects. The mixed effect structure was the maximal one in the sense of Barr et al. 2013. Omitting the numerical details, it revealed effects of PICTURE and POLARITY, but crucially no effect of, or interaction with, CONTEXT. This suggests that when analysing the TARGET conditions, we can confidently consider any effect of CONTEXT to be specific to the TARGET conditions, rather than an overall artifact of the task, although an effect of POLARITY, if any, may not be straightforwardly interpretable.

The data from the TARGET condition were analysed in a similar way to Experiment 1, with a proportional-odds cumulative logit mixed effects model with the same variable coding (i.e., sum-coding for CONTEXT with STRICT as -0.5 and LAX as 0.5 and treatment-coding for POLARITY with NEGATIVE as reference) and the same fixed effect structure. The random effects consisted of by-participant variance for POLARITY and by-item variance for the intercept. We left out by-participant variance for the intercept, as it shows a high collinearity with the by-participant random slope (with $\rho = -0.968$). Also, as in the case of Experiment 1, the by-item

Fixed effects	β	s.e.	p
θ_1	-1.88	0.10	
θ_2	-0.60	0.09	
θ_3	0.20	0.09	
θ_4	2.15	0.10	
CONTEXT	3.26	0.17	< 0.001***
POLARITY	0.13	0.17	0.47
CONTEXT $\times$ POLARITY	-0.17	0.35	0.62
Random effects	σ^2	σ	
By-participant POLARITY	3.33	1.83	
By-item intercept	< 0.001	< 0.001	

Table 8 Estimates of the cumulative logit mixed effects model for Experiment 2. The p -values were obtained via likelihood ratio tests.

random intercept does not explain additional variance in the data and leaving it out would make no difference, but it is kept here.

The estimates of the model are summarised in Table 8. We observe a simple effect of CONTEXT (*not every* received a higher score in the LAX context; $\chi^2(1) = 97.3, p < .001$), no main effect of POLARITY (no evidence for difference between *every* and *not every* in the TARGET condition, but see the caveat above; $\chi^2(1) = 0.53, p = 0.469$), and no significant interaction between CONTEXT and POLARITY ($\chi^2(1) = 0.24, p = 0.624$).

3.4.4 Discussion

In Experiment 2, like in Experiment 1, we found a clear effect of CONTEXT. Importantly, however, this time we failed to observe a significant interaction effect between CONTEXT and POLARITY. The results therefore suggest that in this case CONTEXT had the same degree of robust effect on the two quantifiers *every* and *not every*.

The symmetry in the contextual effect and the overall similar acceptance of the positive and negative quantifiers is very much in line with the symmetric approach, and challenging for the asymmetric approach. Moreover, the difference from Experiment 1 is an issue for both approaches. We will discuss this in more detail in Section 4.

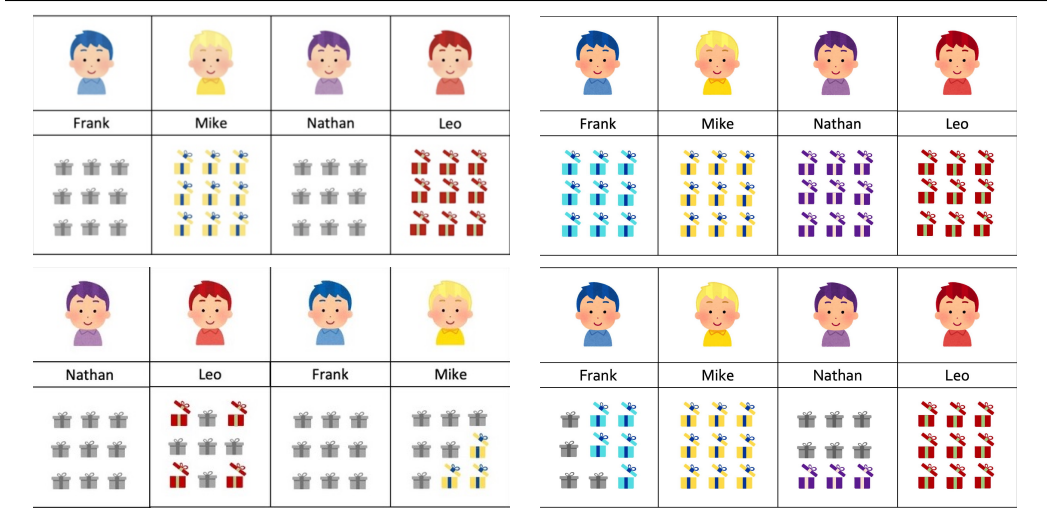


Figure 7 Examples of pictures used in Experiment 3. The top left picture is a TRUE CONTROL item and the top right picture is a FALSE CONTROL item. The bottom left picture was a target item for POSITIVE and the bottom right picture was a target item for NEGATIVE.

3.5 Experiment 3

Experiment 3 investigated the interpretation of plural definites in the scope of the non-monotonic quantifier *exactly two*. The theoretical predictions, as we discussed, diverge here as well. As we will see, while the results are more in line with the symmetric approach, they have significant implications for both approaches.

3.5.1 Methods

The overall design of Experiment 3 is essentially the same as the previous two experiments, except that the target sentence was always (34). As discussed, (34) is non-monotonic, and could be described as being both positive and negative, in some sense, which allows us to investigate the issue of polarity within the same sentence.

(34) Exactly two boys/girls opened their presents. NON-MONOTONIC

The sentence was presented with one of four types of pictures, TRUE CONTROL, FALSE CONTROL, and two TARGET conditions, as illustrated in Figure 7. The idea is that in the POSITIVE TARGET condition, we tested if *their presents* can be read non-maximally with respect to the positive part of the meaning, while in the NEGATIVE

SENTENCE	CONTEXT	
	PROHIBITED	REQUIRED
Exactly two boys opened their presents (<i>positive part</i>)	✓	×
Exactly two boys opened their presents (<i>negative part</i>)	×	☒

Table 9 Predictions of the symmetric approach.

SENTENCE	CONTEXT	
	PROHIBITED	REQUIRED
Exactly two boys opened their presents (<i>positive part</i>)	✓	×
Exactly two boys opened their presents (<i>negative part</i>)	×	☐

Table 10 Predictions of the asymmetric approach.

TARGET condition, we tested if *their presents* can be read non-maximally with respect to the negative part of the meaning.

As in the previous two experiments, the experimental sentences were spread over 8 lists (2 contexts $\times$ 2 genders $\times$ 2 ways of labeling the response buttons). Each list contained 4 items in each of the four experimental conditions, so 16 items in total. All of these items contained the same sentence in (34).

192 native speakers of English were recruited on Prolific (<https://www.prolific.co>), and paid £1.05 for their participation. Using the same exclusion criterion as before (the participants whose accuracy rate on the control items at or below 75% were excluded from the analysis), we are left with 180 participants, whose average accuracy rate on the control items is 98.3%.

3.5.2 Predictions

Recall that the asymmetric approach predicts an effect of context on the availability of the non-maximal readings only in the positive part of *exactly* but not in the negative part. Conversely, the symmetric approach predicts a similar effect of context in the two parts of the meaning. Tables 9 and 10 summarize the relevant predictions again.

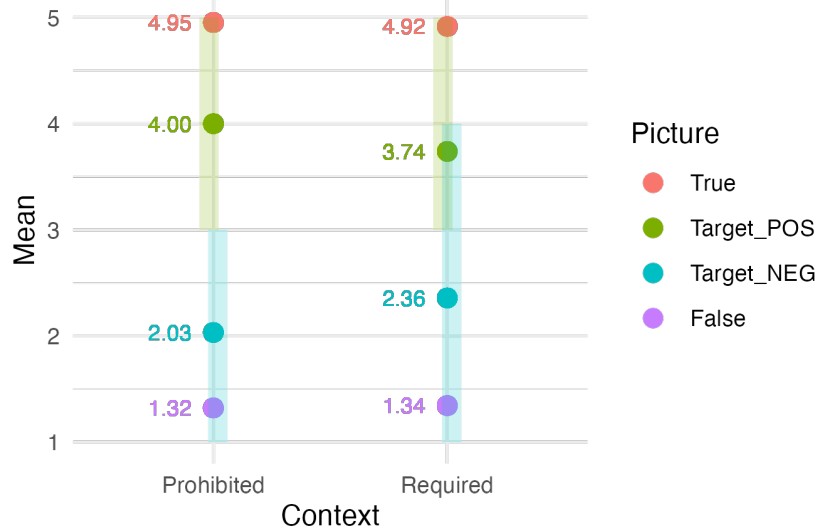


Figure 8 Mean ratings in Experiment 3. Bars indicate interquartile range.

3.5.3 Results

Figure 8 shows the mean ratings for the experimental conditions. Note that the contexts are not recoded in this figure, as the control items are shared between the two target conditions in this case, but in the statistical analysis reported below were conducted with recoding, similarly to the previous two experiments. The two categorical variables, CONTEXT and POLARITY, were coded in the same way as in the previous experiments, i.e., sum-coding for CONTEXT (STRICT as -0.5 and LAX as 0.5) and treatment-coding for POLARITY with NEGATIVE as the reference level.

We fitted a proportional-odds cumulative logit mixed effects model to the data from the TARGET condition, using `clmm()`. This time, the model with a random by-participant intercept did not converge, so we did not include it.

The model estimates are summarised in Table 11. As before, the p -values were obtained via likelihood ratio tests with simpler models using `anova()`. We observe a significant simple effect of CONTEXT (CONTEXT had a significant effect on NEGATIVE; $\chi^2(1) = 15.8$, $p < 0.001$), and a significant main effect of POLARITY (POSITIVE was on average judged as more true than NEGATIVE; $\chi^2(1) = 34.2$, $p < 0.001$), but no significant interaction effect ($\chi^2(1) = 0.019$, $p = 0.892$).

Fixed effects	β	s.e.	p
θ_1	-0.11	0.26	
θ_2	0.73	0.26	
θ_3	1.50	0.27	
θ_4	2.93	0.28	
CONTEXT	0.61	0.15	< 0.001***
POLARITY	3.15	0.41	< 0.001***
CONTEXT $\times$ POLARITY	0.06	0.41	0.89
Random effects	σ^2	σ	
By-participant POLARITY	4.85	2.20	
By-item intercept	0.98	0.99	

Table 11 Estimates of the cumulative logit mixed effects model for Experiment 3. The p -values were obtained via likelihood ratio tests.

3.5.4 Discussion

To summarize the main findings, we observe that non-maximality in the negative aspect of the interpretation of non-monotonic sentences containing plural definites is modulated by context and that non-maximality in the positive aspect of the meaning received higher ratings than non-maximality in the negative aspect of the meaning. We should also note that the effect size of context manipulation is much smaller than in Experiment 2, and is numerically even smaller than the simple effect on *no* in Experiment 1.

The lack of a significant interaction effect suggests that non-maximality in the two aspects of the meaning of definite descriptions under *exactly two* is roughly equally context-sensitive. This is compatible with the predictions of the symmetric approach, but challenging for the asymmetric approach.

In addition, the difference in effect size from the previous experiment, especially the fact that among the negative contexts the context manipulation had the largest effect with *not every*, is puzzling for both approaches. We will come back to these points with more details in the next section.

4 General discussion

4.1 Main results

Let us first summarize the main experimental findings.

Quantifier	Likert scale	Log-odds
Every (Exp. 1)	1.13	3.84
Every (Exp. 2)	1.60	3.09
Not every	1.91	3.26
No	0.50	1.41
Exactly two (positive)	0.26	0.66
Exactly two (negative)	0.33	0.61

Table 12 Effect size of context manipulation in the experimental conditions of the three experiments. The effect sizes on the Likert scale are the differences between mean ratings, and the effect sizes in terms of log-odds are computed from the regression models’ fixed effects estimates.

- We found evidence for an asymmetric effect of our context manipulation for non-maximal readings of plural definites in the scope of *every* and *no*, but did not find the same evidence in Experiment 2 and Experiment 3.
- In Experiment 1 and Experiment 3 the negative condition received much lower ratings than the positive one.
- In all three experiments, context-sensitivity was observed in the negative condition, as well as in the positive condition.
- In addition, variation in the effect size of context manipulation was observed. That is, a relatively large effect was observed between the two types of context for plural definites under *every* and *not every* but the effect was much smaller under *no* and *exactly two* (see Table 12).

We will discuss the challenges these findings pose for the symmetric and asymmetric approaches in detail, and propose possible auxiliary assumptions and hypotheses that each approach could adopt to account for our results.

4.2 The symmetric approach

The main challenges for the symmetric approach are as follows. First, *no* gave rise to non-maximal readings much less often than both *every* and *not every*. This pattern calls for explanation, since the symmetric approach predicts that definite plurals should be equally sensitive to context, regardless of the semantic properties of the grammatical environments in which they occur. Second, the difference between the positive and negative conditions in Experiments 1 and 3 is likewise unexpected under

this approach. Finally, the variation in effect size across the three experiments also requires an explanation.

We consider here a hypothesis that would bring the symmetric approach into closer alignment with our experimental results.¹² The core idea is that our direct manipulation of context allowed us to control its effects across different quantifiers and to show that context has a robust influence on the availability of non-maximal readings. Crucially, however, this influence appears to vary depending on the quantifier involved.

While this poses a challenge for the symmetric approach, we cannot rule out the possibility that the effect size of context manipulation, as well as the baseline from which it starts, is negatively correlated with the strength of prior associations between particular sentence types and particular kinds of contexts. More specifically, with respect to Experiment 1, it may be that sentences containing *no* are associated with a prior biased toward an existential QUD, e.g., whether any boy opened any of his presents. Such a prior would both attenuate the effect of our contextual manipulation and reduce the availability of non-maximal readings of plural definites in its scope, relative to sentences containing *every* or *not every*.

The results of Experiment 3 are also compatible with this hypothesis. Given the nature of non-monotonic quantifiers, these sentences allowed us to test the availability of non-maximal readings in both the positive and negative components of a single sentence. In other words, because we were examining the positive and negative entailments of the same sentence, symmetric behavior is in fact expected, and this is what we observed. In addition, assuming a sufficiently skewed prior, the hypothesis is also consistent with the smaller effect size found in this experiment. Finally, it can also accommodate the main effect observed between the two components. On the assumption that the priors associated with the non-monotonic sentences are biased toward existential QUDs, it is expected that the positive condition, in which two children opened some but not all of their presents and the other two opened none, should receive higher ratings than the negative condition, in which all four children opened some or all of their presents.

Having said this, the hypothesis is at present no more than a promissory note for the symmetric approach. For it to have genuine explanatory force, the priors required to account for the data under the symmetric approach would need independent support. At this stage, however, it is not clear why a sentence with *no* should be associated with the QUD described above rather than, say, with a universal QUD or some other QUD that would be incompatible with the pattern observed in our results. We therefore leave further exploration of this hypothesis for future work.

¹² We are grateful to an anonymous reviewer who helped us sharpen this hypothesis significantly.

4.3 The asymmetric approach

There are three main challenges for the asymmetric approach. First, the observed effect of context for negative sentences, both with *no* and *not every*, is unexpected, at least under the version of the theory sketched above (but see below). Second, it is unclear why, on this approach, definite plurals under *not every* gave rise to non-maximal readings to the same extent as those under *every*, and were equally sensitive to contextual effects, in contrast to *no*. Similarly, the negative component of *exactly two* allowed non-maximal readings to the same extent as the positive component, contrary to the predictions of the asymmetric approach. Finally, the effect size in Experiment 3 was substantially smaller than in Experiments 1 and 2.

Building on observations by Bar-Lev 2021, Guerrini & Wehbe (2025) acknowledge the first of these challenges on the basis of introspective judgments and suggest that non-maximal readings in negative sentences are possible, albeit at a cost. The proposal is that, although accommodation of the implicature is preferentially global, it may also occur locally under negation, as in (35). This introduces the homogeneity implicature within the scope of the quantifier, thereby weakening the overall meaning and making it compatible with non-maximal scenarios. The resulting meaning is roughly paraphrasable as: no boy opened all of his presents.

$$(35) \quad \llbracket [\text{no boy}]_2 A[\text{PEX}[t_2 [\text{his presents}]_1 [\exists\text{-PLD opened } t_1]]] \rrbracket = \\ \neg \exists x [\text{boy}(x) \wedge \forall y \in \{a, b, c\} [\text{present.of } x(y) \rightarrow \text{opened}(x, y)]]$$

Importantly, as is standardly assumed for local accommodation, this option is available but dispreferred and costly, which provides a natural explanation for the asymmetry between *every* and *no*.

Having said that, the symmetric patterns observed in Experiments 2 and 3 remain puzzling for this account, since *not every* and the negative component of *exactly two* are predicted to pattern with *no*. One might consider appealing here as well to a local accommodation explanation, which could in principle give rise to a weaker reading, at least in the *not every* case.¹³ The difficulty, however, is that, as just mentioned, local accommodation is crucially assumed to be costly, in order to preserve the asymmetry between *every* and *no*. As a result, even if the option of local accommodation is considered, the approach remains incompatible with the symmetry observed in Experiments 2 and 3.

Finally, the asymmetric approach does not straightforwardly explain why the effect of contextual manipulation was substantially weaker in Experiment 3 than in Experiments 1 and 2.

¹³ For the non-monotonic cases, this idea could in principle be extended by assuming a compositional analysis of the positive and negative components. We do not pursue this possibility here, given the independent problem for this line of explanation discussed in the main text.

In sum, while the implicature theory can be made fully compatible with our results of Experiment 1, some aspects of the results of Experiment 2 and 3 and their differences remain challenging.

5 Conclusion

We reported on three experiments using a picture-sentence truth-value judgment task, investigating sentences with plural definites in the scope of positive, negative, and non-monotonic quantifiers, against different contexts. The experiments were designed to test divergent predictions of two main approaches in the literature with respect to non-maximal readings of plural definites: the symmetric and asymmetric approaches. Our results provided evidence for an asymmetry between plural definites in the scope of *every* and *no* sentences but a symmetry between plural definites in the scope of *every* and *not every*, and the positive and negative part of non-monotonic quantifiers such as *exactly two*. In addition, we found that non-maximal readings were possible and affected by context in all of these environments, and that the effect of context was larger in some cases than others. We argued that taken together, these results are challenging for both approaches to non-maximal readings of plural definites, and discussed possible directions each approach can explore in order to account for our results and the subsequent predictions these theoretical choices make, to be tested in future work.

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